ELSEVIER

Atmospheric Environment 38 (2004) 3329–3340

ATMOSPHERIC ENVIRONMENT

www.elsevier.com/locate/atmosenv

# Performance evaluation of the active-flow personal DataRAM $PM_{2.5}$ mass monitor (Thermo Anderson pDR-1200) designed for continuous personal exposure measurements

Bhabesh Chakrabarti[a], Philip M. Fine[a], Ralph Delfino[b], Constantinos Sioutas[a,*]

[a] *Department of Civil and Environmental Engineering, School of Engineering, University of Southern California, 3620 South Vermont Avenue, Los Angeles, CA 90089, USA*

[b] *Epidemiology Division, Department of Medicine, University of California, Irvine School of Medicine, 224 Irvine Hall, Irvine, CA 92697-7550, USA*

**Abstract**

The need for continuous personal monitoring for exposure to particulate matter has been demonstrated by recent health studies showing effects of PM exposure on time scales of less than a few hours. Filter-based methods cannot measure this short-term variation of PM levels, which can be quite significant considering human activity patterns. The goal of this study was to evaluate the active-flow personal DataRAM for $PM_{2.5}$ (MIE pDR-1200; Thermo Electron Corp., Franklin, MA) designed as a wearable monitor to continuously measure particle exposure. The instrument precision was found to be good (2.1%) and significantly higher than the passive pDR configuration tested previously. A comparison to other proven continuous monitors resulted in good agreement at low relative humidities. Results at higher humidity followed predictable trends and provided a correction scheme that improved the accuracy of pDR readings. The pDR response to particle size also corresponded to previously observed and theoretical errors. The active flow feature of the pDR allows collection of the sampled particles on a back-up filter. The 24-h mass measured on this filter was found to compare very well with a Federal Reference Method for $PM_{2.5}$ mass.
© 2004 Elsevier Ltd. All rights reserved.

*Keywords:* Personal exposure; Personal monitor; $PM_{2.5}$; DataRAM; pDR; PM; Particulate matter; Nephelometer

## 1. Introduction

Exposure to ambient particulate matter (PM) has recently received considerable attention as the result of epidemiological findings showing associations between ambient PM, $PM_{10}$ and $PM_{2.5}$ concentrations (particles with diameters less than 10 and 2.5 μm, respectively) and mortality (Ozkaynak and Thurston, 1987; Schwartz and Dockery, 1992; Dockery et al., 1993; Pope III et al., 1991; Katsouyanni et al., 1995; Pope et al., 1995; Schwartz et al., 2002). Findings from these health studies have raised concerns about the sufficiency of the National Ambient Air Quality Standard (NAAQS) for PM, in terms of the regulated level, the upper size cut of the particles, the adequacy of measuring only particle mass, and the applicability of fixed-site monitoring to personal exposure. These concerns have been difficult to address, since little is known about the relationship between outdoor particle concentrations and actual human exposure. Several studies have indicated that outdoor and even indoor concentrations may be poor estimators of personal exposures to $PM_{10}$ or $PM_{2.5}$ and its components. For example, daytime personal $PM_{10}$ exposures were found to be approximately 50% higher than corresponding indoor and outdoor levels (Thomas et al., 1993), while personal $SO_4^{2-}$ and $H^+$ exposures were found to be higher than indoor, but lower than outdoor concentrations (Suh et al., 1994).

*Corresponding author. Tel.: +1-213-740-6134; fax: +1-213-744-1426.

*E-mail address:* sioutos@usc.edu (C. Sioutas).

1352-2310/$ - see front matter © 2004 Elsevier Ltd. All rights reserved.
doi:10.1016/j.atmosenv.2004.03.007

Understanding patterns of individual exposures to PM can be significantly improved by the use of personal monitors, as these samplers incorporate the effects of factors such as indoor pollutant sources and human activity patterns.

A variety of personal samplers have been developed to-date. Amongst those commercially available are: the $PM_{2.5}$ Personal Exposure Monitor (PEM Model 200 MSP Corp., Minneapolis, MN, 4 liters per minute ($l/min^{-1}$)) (Buckley et al., 1991); a personal $PM_{2.5}$ sampler with cyclone (GK2.05 KTL Cyclone, BGI Inc., Waltham, MA, $4 l/min^{-1}$) used in the European 6-city EXPOLIS study and others (Koistinen et al., 1999); the Institute of Occupational Medicine sampler (IOM Personal Inhalable Dust Sampler, SKC inc., Eighty Four, PA, $PM_{10}$) (Mark et al., 1986); a High Flow Personal Sampler (HFPS) for $PM_{2.5}$ (Adams et al., 2001) with a flow rate of $16 l/min^{-1}$ that allows for shorter sampling periods; and a Personal Cascade Impactor Sampler (PCIS) that provides size-fractionated PM personal data at $9 l/min^{-1}$ (Misra et al., 2002). In all of the aforementioned personal samplers, particle mass concentrations are determined based on gravimetric analysis of filter and/or impaction substrates on which PM are collected over periods of several hours or an entire day. Although federal air quality standards for PM in the United States are based on integrated 24-h and annual mass concentrations, recent studies have provided evidence of adverse health effects from short-term exposures to PM (US EPA, 1996; Schwartz and Neas, 2000; Delfino et al., 1998; Peters et al., 2001). Nevertheless, the values of key metrics influencing personal exposure to PM, such as the emission strengths of outdoor and indoor particle sources, ''personal cloud'' generation, and other micro-environmental factors, fluctuate on time scales that are substantially shorter than several hours. Individual activity patterns influencing exposure, such as amount of time spent indoors, outdoors and in commute, also vary on time periods considerably shorter than several hours.

Several recent articles have hypothesized about the health significance of brief airborne particle excursions on a time scale of a few minutes to several hours, suggesting that these excursions might explain some of the excess mortality (Michaels, 1996, 1997; Michaels and Kleinman, 2000). This hypothesis is based on a number of acute human exposure studies ($\sim$25) cited in Michaels and Kleinman (2000) and recent animal studies (Godleski et al., 1999; Clarke et al., 2000) that show evidence of adverse health effects due to such brief excursions. Two recent asthma panel studies of 22–24 asthmatic children examined effects of hourly PM10 data and found maximum 1-h PM10 showed the strongest associations with asthma symptoms (Delfino et al., 1998, 2002). The need for developing personal monitors that measure particle concentration in shorter time intervals (on the order of 1–2 h, or less) is therefore of paramount importance to environmental health, as it leads to substantial improvements in exposure assessment to ambient particulates.

The objective of the study presented in this paper was to evaluate the performance of the active-flow, personal DataRAM (MIE pDR-1200; Thermo Electron Corp., Franklin, MA) for providing real-time mass concentration measurements and to assess the effect of particle size and humidity on the relationship between the response of the pDR and the actual aerosol mass concentration. The active pDR, like previous DataRAM models (MIE pDR-1000AN; Thermo Electron Corp., Franklin, MA) (Liu et al., 2002), is an integrating nephelometer that measures continuously the amount of light (with wavelength, $\lambda$, equal to 880 nm) scattered by particles drawn though a sensing zone at a flow rate of $4 l/min^{-1}$. The amount of light scattered is converted to particle concentration readings using well-established light scattering theory (Kerker, 1969) and factory calibrations. In this paper, we present results from a field evaluation of this instrument conducted in Southern California from 6 December, 2002 to 19 February, 2003. The pDRs were tested while stationary (not worn by subjects for personal monitoring) and thus, the current study addresses the accuracy and precision of the instrument's measurement technique, but does not address reliability or durability issues that may be encountered in actual field use. The performance of the active pDR was compared to other collocated continuous and time-integrated $PM_{2.5}$ samplers, which is not practical for actual personal sampling.

## 2. Methods

### 2.1. Thermo MIE personal DataRAM

The active pDR (shown in Fig. 1) is an integrating nephelometer (scattering coefficient range $= 1.5 \times 10^{-6}$–$0.6 m^{-1}$ at wavelength 880 nm). It has maximum external dimensions of 153 mm (6.0 in) $H \times 92$ mm (3.6 in) $W \times 63$ mm (2.5 in) $D$, weighs 0.5 kg and operates in a temperature range of $-20$–$70^{\circ}C$ and relative humidity range from 10% to 95%. It records in units of $\mu g m^{-3}$, as calibrated by the manufacturer using a fine ISO test dust (specific gravity, 2.5–2.6; MMD, 2–3 μm). For this study, the pDRs were programmed to record 15 min averages. The pDR is preceded by a metal cyclone (BGI model GK 2.05, Waltham MA, provided by Thermo Electron Corp.) with a cut point of 2.5 μm at $4 l/min^{-1}$, designed specifically for $PM_{2.5}$ monitoring. An advantage of using the pDR in the active mode is the ability to easily attach a filter holder for 37 mm filters directly

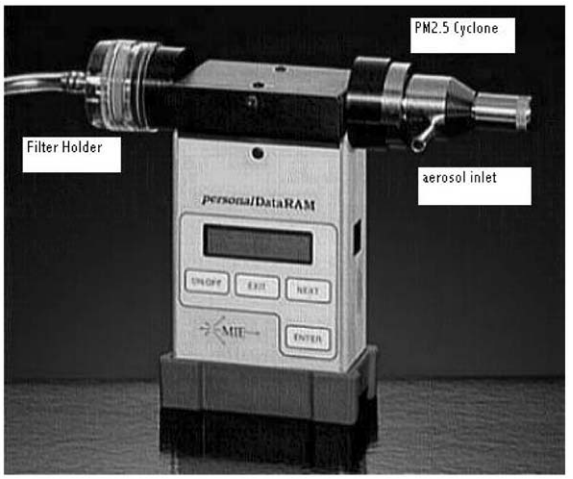



Fig. 1. The personal Data RAM-1200.

downstream of the photometric sensing chamber. The filter can then be analyzed for mass or chemical composition much like the time-integrated personal monitors mentioned in the previous section. In the current experiment, Teflon membrane filters (PTFE, 2 μm pore, Gelman Science, An Arbor, MI) were deployed for concurrent gravimetric measurements of the aerosol mass concentration for comparison to the continuous light-scattering measurements provided by the instrument. Air samples were drawn through each pDR by means of a lightweight, personal pump (BGI 400S Personal Sampling Pump, Waltham, MA).

### 2.2. Field study set-up

Three identical active pDR units with corresponding personal pumps were placed inside an instrumentation trailer located near the University of Southern California in downtown Los Angeles as part of the activities of the Southern California Particle Center and Supersite. In this current setup, outdoor air is first pumped through a Model TE-6001 Size Selective Inlet (Tisch Environmental Inc., Cleveland, OH), similar to those used in high-volume air samplers for routine monitoring. The ambient aerosol then is drawn through the roof of the trailer via a stainless-steel tube which then branches into three sampling lines, each leading to one of three pDR units for simultaneous outdoor air sampling. Each of the three pDRs operates at a flow of $4\,l/min^{-1}$ and is preceded by a $PM_{2.5}$ cyclone described above. Before each experiment, the DataRAM was calibrated following standard calibration procedures recommended by the manufacturer. These include zeroing of the instrument (i.e., testing with particle-free air provided by a filter supplied by the manufacturer) and span checking (i.e., a secondary calibration which is performed using a built-in optical scattering/diffusing element). This internal calibration ensures that before each test started, the calibration factor was reset to the manufacturer's laboratory calibration. This factor can be adjusted by the user to account for differences between the calibration dust used by the manufacturer and ambient aerosol. For the collocated precision tests, the pDRs were not zeroed regularly in order to simulate field operations where frequent zeroing may not be possible. However, all tests with filters were preceded by the zeroing procedure.

For comparison purposes, 1-min continuous pDR data was converted to 15-min, 2-h or 24-h averages according to the sampling periods of the other samplers listed below. The 15-min average measurements of the three collocated pDR measurements were compared. If two of the pDRs agreed within $1\,\mu g\,m^{-3}$, and the third differed by more than 25%, the data from the differing pDR was excluded from the analysis. Less than 3% of the 15-min average data met these criteria and were excluded.

Time-integrated $PM_{2.5}$ sampling was conducted in simultaneous 24-h runs using a Partisol sampler (Model 2025 Sequential Air Sampler, Rupprecht and Patashnick Co. Inc., Albany, NY; Patashnick et al., 2001), designated as a Federal Reference Method (FRM) for $PM_{2.5}$, and the pDR back-up filter. As the pDRs were operating continuously, Teflon filters for mass measurements were deployed in both the FRM sampler and the pDR back-up filter holder for a total of twenty 24-h periods (9 AM to 9 AM). Teflon filters from both the FRM and pDR were pre- and post-weighed using an MT5 Microbalance (Mettler Toledo Inc., Highstown, NJ). Relative humidity was measured every 15 min by a thermo hygrometer HI-9161F (Hanna Instruments, Italy).

The pDR continuous $PM_{2.5}$ concentrations are compared with those measured by a collocated Beta Attenuation Monitor (BAM, Model 1020, Met One instruments, Inc., OR), preceded by a size selective inlet of $PM_{2.5}$. The BAM has been designated as a United States Environmental Protection Agency Equivalent Method for 24-h $PM_{10}$. In the current study, the BAM was considered as the reference for continuous $PM_{2.5}$ measurements over other continuous monitors such as the Tapered Element Oscillating Microbalance (TEOM™). Previous field investigations in California (Chung et al., 2001) showed that performance of the BAM was in excellent agreement with the $PM_{2.5}$ filter-based Federal Reference Method ($R^2 = 0.99$), was not affected by particle composition or meteorological conditions, outperformed TEOMs and nephelometers, and thus appeared to be a suitable candidate for deployment in a real-time continuous $PM_{2.5}$ monitoring network. In the current study, The BAM monitor was

set to automatically record 2-h averages of $PM_{2.5}$ concentrations. The continuous pDR concentrations were also compared with 24-h time-integrated data determined gravimetrically from the pDR back-up filter and the Partisol sampler.

The size distribution of ambient aerosols, which is expected to affect the response of any nephelometer such as the pDR (as discussed in the following section), was measured near-continuously during the field experiments by means of a TSI Scanning Mobility Particle Sizer (SMPS Model 3936) and Aerodynamic Particle Sizer (APS Model 3021, TSI Inc., Shoreview, MN). The size distribution for each 15-min pDR cycle was characterized by a single parameter, the mass median diameter (MMD). The MMD is the particle diameter at which 50% of the particle mass is associated with particles less than (and greater than) this diameter. The calculation of MMD was performed using a simple spreadsheet application.

## 3. Results and discussion

### 3.1. Precision of the active pDR monitors

The precision of the collocated pDR monitors for measurement of the ambient $PM_{2.5}$ mass concentration was found to be very high. Figs. 2a and b illustrate the degree of correlation found among the 15-min average readings of three collocated pDRs. The three pDR readings were highly correlated with $R^2$ greater than 0.99 and slopes within $\pm 10\%$ of unity in all cases. In our collocated precision tests, the pDRs were not necessarily zeroed regularly in order to more closely approximate field operations where frequent zeroing may not be practical. The geometric mean ratios near unity and the small mean relative differences shown in Table 1 also confirm the excellent agreement among the three pDR. Due to the high level of precision among the pDR monitors, the average of the three pDR readings is used

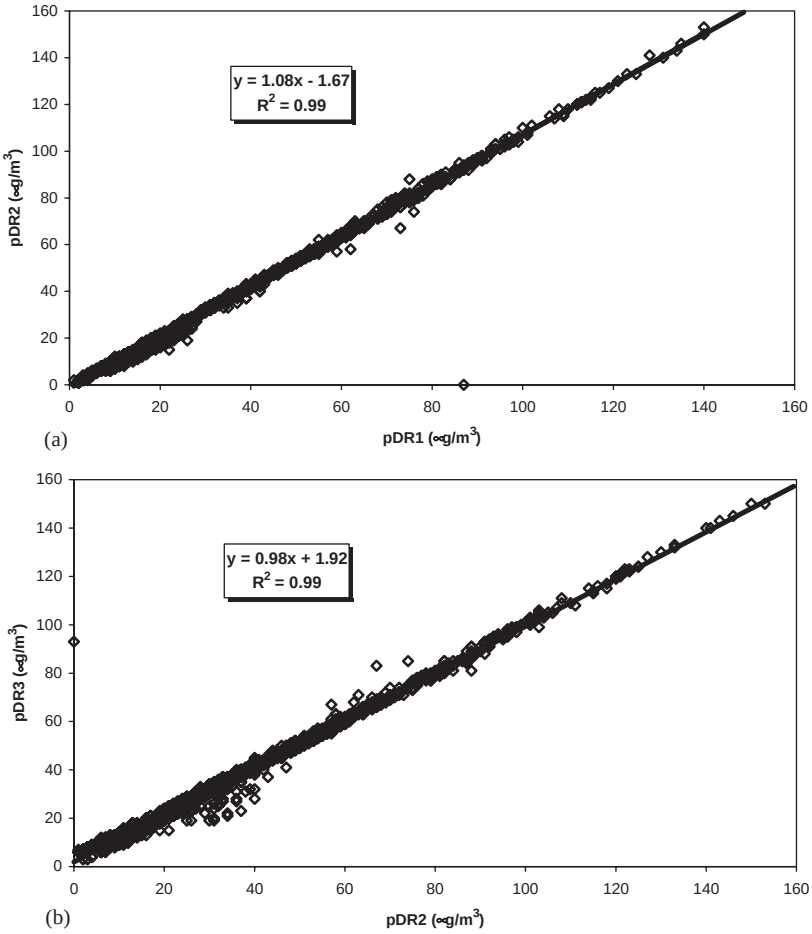



Fig. 2. Correlation of 15-min average $PM_{2.5}$ concentrations between (a) pDR1 and pDR2 and (b) pDR2 and pDR3.

Table 1
Summary statistics of the comparisons between the $PM_{2.5}$ concentrations obtained by pDR, BAM and Partisol

| Instruments compared | *N* | Geometric mean of ratios | Mean relative difference[a] | *p*-Value[b] |
|---|---|---|---|---|
| *Continuous comparison – precision among pDRs* (15 *min*) | | | | |
| pDR1 vs. pDR2 | 573 | 1.029 | 0.133 | 0.299 |
| pDR1 vs. pDR3 | 559 | 1.023 | 0.135 | 0.163 |
| pDR2 vs. pDR3 | 564 | 1.015 | 0.089 | 0.323 |
| *Semi-continuous comparison – pDR* vs. *BAM* (2-*h average*) | | | | |
| Average pDR vs. BAM (All RH) (2 h) | 206 | 1.33 | 0.165 | $<0.001$ |
| Average RH-Corrected pDR vs. BAM (RH > 60%)(2 h) | 111 | 0.97 | 0.189 | 0.31 |
| Average pDR vs. BAM (RH < 60%)(2 h) | 95 | 0.98 | 0.161 | 0.11 |
| Gravimetric comparison (24 h) | | | | |
| Gravimetric[c] pDR1 vs. pDR2 | 20 | 1.027 | 0.071 | 0.367 |
| Gravimetric[c] pDR1 vs. pDR3 | 20 | 1.004 | 0.080 | 0.387 |
| Gravimetric[c] pDR2 vs. pDR3 | 20 | 1.032 | 0.105 | 0.485 |
| Average gravimetric[c] pDR vs. partisol | 20 | 1.027 | 0.117 | 0.391 |
| Average gravimetric[c] pDR vs. BAM | 20 | 1.112 | 0.170 | 0.089 |
| Average light-scattering pDR vs. Average gravimetric[c] pDR | 20 | 1.528 | 0.79 | 0.001 |
| Average RH-Corr. light-scattering pDR vs. average gravimetric[c] pDR | 18 | 1.058 | 0.325 | 0.129 |
| Average BAM vs. partisol | 20 | 0.92 | 0.13 | 0.11 |

[a] The relative difference is defined as the absolute difference between the measurements divided by the reference measurement. The mean relative difference is the average of the relative differences.
[b] Based on a paired-sample *t*-test.
[c] Gravimetric pDR measurements are based on the change in weight of the back-up Teflon filter.

for all subsequent comparisons to other particle measurements.

A previous study on the performance of the passive (no active flow) configuration of similar pDRs was conducted by Liu et al. (2002). Among the findings of that study was the increase in the degree of precision as PM concentration increased. Figs. 2a and b graphically demonstrate this effect with less scatter seen at the higher concentrations. The Liu et al. study found precisions in the range of 5–24% (defined as the square root of the average variance among instruments divided by the number of collocated instruments, and then divided by the mean measurement), depending on particle concentration. The precision in the current study, calculated in the same manner but over all ranges of particle concentrations, was 2.1% suggesting that the active flow configuration significantly increases monitor precision.

### 3.2. *pDR/BAM inter-comparison and humidity effects*

Table 1 also presents results of the inter-comparison between the average of the three pDR readings (averaged over 2-h time periods) and the 2-h BAM monitor output, which we use as a reference for $PM_{2.5}$ concentrations. With a geometric mean pDR to BAM ratio of 1.33, the pDR reads significantly ($p<0.001$) higher particle levels than the BAM. For the subset of data corresponding to ambient relative humidities exceeding 60%, the pDR to BAM concentration ratio increases to 1.71. Previous studies have shown that light-scattering particle measurement devices are subject to error at high relative humidities (Sioutas et al., 2000; Day et al., 2000). A previous evaluation of a stationary DataRAM (also manufactured by Mie, Inc.) showed that the DataRAM technology overestimates particle concentrations at relative humidities (*RH*) above approximately 60% (Sioutas et al., 2000). Fig. 3 plots the pDR/BAM ratio as a function of 2-h average relative humidity. The results are very similar to the previously reported stationary DataRAM performance (Sioutas et al., 2000) with the ratio increasing almost exponentially above an RH of 60%. The model fit included on Fig. 3 is based on an empirical correction factor (CF) described by the

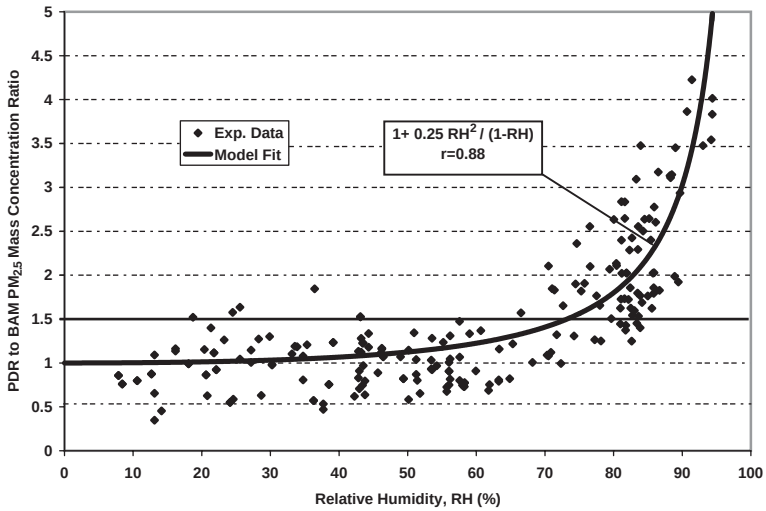



Fig. 3. Two-hour averaged pDR to BAM $PM_{2.5}$ concentration ratio as a function of relative humidity.

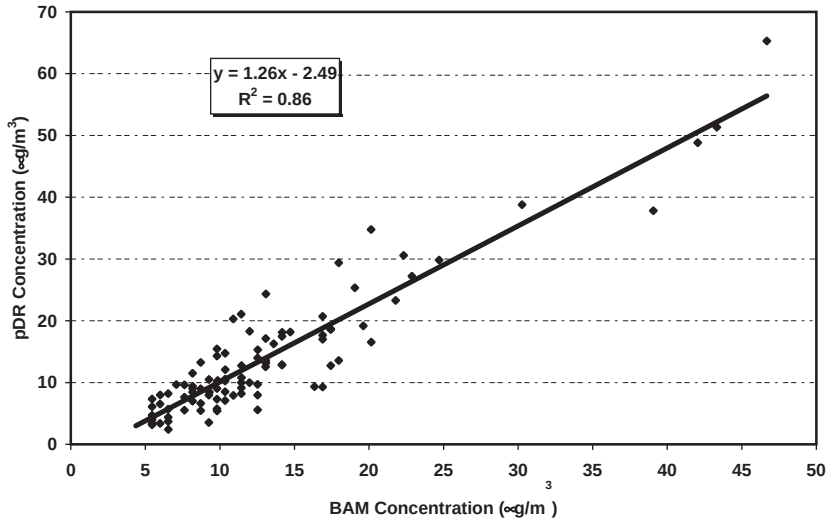



Fig. 4. Correlation between 2-h averaged pDR and BAM $PM_{2.5}$ mass concentrations (relative humidity $<60\%$).

Laulainen (1993):

$$CF = 1 + 0.25RH^2/(1 - RH), \quad (1)$$

where RH is the relative humidity. This model was also shown to fit more recent data presented by Day et al. (2000) quite well. As Fig. 3 indicates, the modeled particle growth predicted by that equation fits the actual field data well with a correlation coefficient of 0.88.

Since a nephelometer device measures particle mass in situ, rather than after collection on a filter material, the measurement will include increased particle mass due to condensational growth of the particles associated with water uptake by hygroscopic PM components (McMurry et al., 1996; Sloane, 1984). The particle mass measured by filter-based methods, which incorporate a filter equilibration step at low RH for time-integrated samples and upstream heaters in the case of the BAM monitor, will not include most of the mass of particle-associated water. Thus, nephelometers will over estimate particle mass at higher relative humidities when condensational particle growth is favored. This effect has been studied and modeled by Sloane (1984) and Lowenthal et al. (1995), with results that resemble the empirical fit of Laulainen (1993).

Fig. 4 graphically displays the relationship between the 2-h average pDR and BAM $PM_{2.5}$ concentrations at

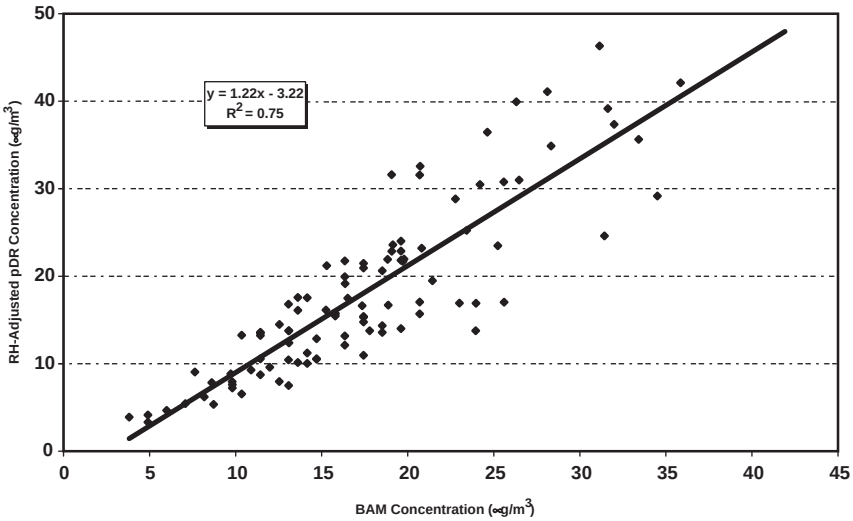



Fig. 5. Correlation between 2-h averaged RH-adjusted pDR and BAM (relative humidity >60%).

times when the RH was less than 60%. High correlation between the measurements is observed ($R^2 = 0.86$). The 2-h averaged $PM_{2.5}$ concentrations determined by pDR and BAM are also in excellent agreement with each other, with the pDR-to-BAM ratio being 0.98, as indicated in Table 1. As already demonstrated in Fig. 3, agreement between pDR and BAM measurements is not expected for RH above 60%. However, by correcting the pDR readings when the RH was greater than 60% with the empirical relationship described in Eq. (1), the two measurements can be brought into reasonable agreement (Fig. 5). Table 1 also demonstrates the effectiveness of this correction, with an uncorrected pDR to BAM ratio of 1.71 (not shown in the table), compared to a corrected concentration ratio of 0.97. A paired *t*-tested showed that the RH-adjusted pDR and BAM concentrations are not statistically different ($p = 0.31$). The scatter of the data plotted in Figs. 4 and 5 suggest the pDR $PM_{2.5}$ concentration may still be somewhat different than that of the BAM, even after adjustment for RH. A possible reason for this observation is the effect of particle size on light-scattering measurements, such as the pDR, which is considered in the following section.

### 3.3. Particle size effects

The effect of the particle size on the pDR response was investigated by calculating the mass median diameter (MMD) as determined by concurrent measurements of size distributions using a tandem SMPS and APS system. The variation of the pDR/BAM ratio as a function of MMD is shown in Fig. 6. In this figure, only time periods when the RH was less than 60% are shown to completely separate the size effects from the biases caused by RH. Certain data points are also excluded when the SMPS/APS system appeared to be malfunctioning (during the period of 2–5 January, 2003). The experimental data follow the classic trend predicted by the Mie scattering theory (Kerker, 1969), with an overestimation of particle mass between about 0.7 and 1.2 μm MMD. The observed pattern, as well as the best fit third order polynomial curve shown in Fig. 6, agrees remarkably well with previously reported results of a stationary DataRAM by Sioutas et al. (2000). It is known that variations in particle size introduce considerable errors in predicting the response of a nephelometer (Sioutas et al., 2000; Thomas and Gebhart, 1994). The intensity of scattered light from an individual particle increases from about 0 (for very small, ultrafine particles) to a maximum value at a particle size that corresponds to approximately 1.25 times the wavelength of the incident light. The pDR uses a light source with wavelength of 0.88 μm, hence maximum scattering should be observed at about 1.1 μm particles. Note that, as shown in Fig. 6, this is the approximate MMD value where the pDR over estimates the BAM measurements to the greatest degree.

Since the pDRs were calibrated using ISO fine test dust (MMD, 2–3 μm; specific gravity, 2.5–2.6, as aerosolized) the instrument response to a finer $PM_{2.5}$ aerosol (smaller particles) will be biased. This response can be theoretically predicted if one knows the particle size distribution and refractive index of the calibration aerosol as well as the test aerosol (Gorner et al., 1995). Since real-time personal monitors for particle size distributions have not yet been developed, correcting for particle size effects is not possible. Therefore, some inherent biases will remain in pDR measurements whenever particle size distributions change over time.

### 3.4. Gravimetric measurements

In addition to evaluating the performance of the pDR on a semi-continuous basis, the gravimetrically determined mass collected on the pDR back-up filter was compared to 24-h time-integrated Federal Reference Methods (FRM) for $PM_{2.5}$. First, the three pDR back-up filters agreed very well with one another with geometric mean ratios ranging from 1.00 to 1.03 as shown in Table 1. Therefore, the average of the three was used for subsequent comparisons. Table 1, along with Fig. 7, also confirm that the pDR filter sampling capability can accurately measures $PM_{2.5}$ mass relative to the FRM (geo. mean ratio = 1.027, $R^2 = 0.93$, slope = 0.91). Fig. 8 shows a comparison between the 24-h average gravimetrically determined pDR concentrations and those measured by the BAM, averaged over 24 h. The 24-h $PM_{2.5}$ concentrations of these two instruments appear to be well correlated ($R^2 = 0.71$) with a gravimetric pDR to BAM geometric mean ratio of 1.11 (Table 1). Although not shown in a figure, the 24-h averaged BAM concentrations are also in very

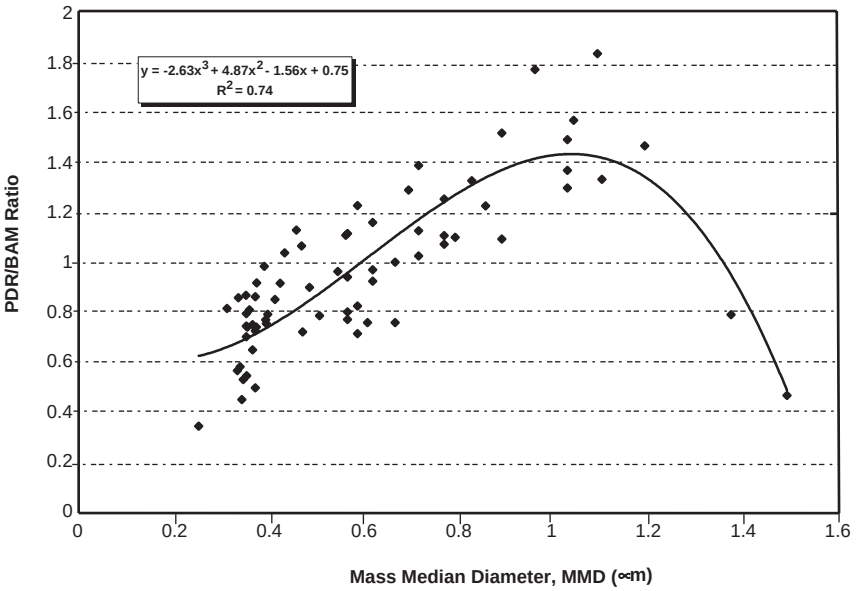



Fig. 6. Two-h averaged pDR to BAM $PM_{2.5}$ concentration ratio as a function of ambient aerosol mass median diameter (MMD).

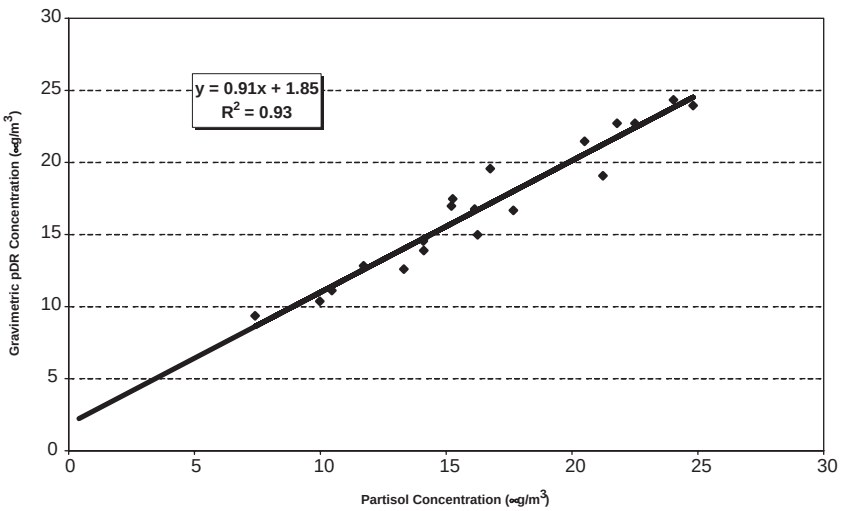



Fig. 7. Correlation between average 24-h gravimetric (back-up filter) pDR and Partisol.

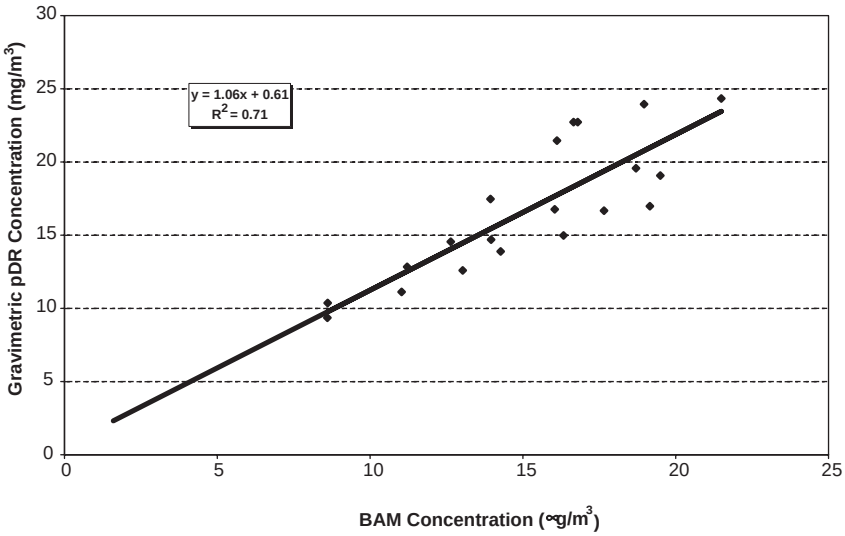



Fig. 8. Correlation between average 24-h gravimetric (back-up filter) pDR and 24-h averaged BAM.

good agreement with those measured with the Partisol, with an average BAM-to-Partisol ratio of 0.92, and an $R^2 = 0.77$. These data are in agreement with the results of the Chung et al. (2001) study and provide further corroboration to the integrity of the performance of the BAM.

Considering the high degree of accuracy with respect to the FRM and other methods, this feature of the pDR allows a direct filter-based measurement of the total time-integrated mass of the same particles continuously monitored by the light-scattering measurements of the pDR. The filter-based measurement might then be used to correct for the known biases of the continuous pDR data. Fig. 9a displays this relationship between the average 24-h gravimetric mass as measured by the three back-up pDR filters and the 24-h average of the continuous data from all three pDRs. The degree of correlation appears to be poor ($R^2 = 0.16$). However, when the humidity correction used above is applied to the continuous pDR on a 15-min basis, prior to 24-h averaging, the correlation improves significantly to $R^2 = 0.56$ (Fig. 9b). Table 1 also demonstrates how using the humidity correction improves the accuracy of the data. The geometric mean ratio between light-scattering and gravimetric data improves from 1.53 to 1.06 with the correction. The remaining scatter after the humidity correction could be due to the particle size biases described previously. An additional correction could hypothetically be applied to account for varying MMD, but in the actual deployment of the pDRs as personal monitors, continuous MMD measurements are not practical. However, continuous recorders of relative humidities are available for personal monitoring, and could thus be deployed to help improve the accuracy of the pDR measurements.

## 4. Summary and conclusions

The field study of the active pDR monitors showed the instruments to be of sufficient precision and accuracy to be useful for personal exposure studies. The precision of the instruments is very high (2.1%) and notably higher than the passive pDR configuration. Comparison to other proven continuous monitors revealed good agreement at lower relative humidities. Results at higher RH followed predictable theoretical and empirical trends that provide an accurate correction of pDR readings if RH is known. The response of the pDRs to particle size was also found to correspond to previously determined theoretical biases, although it is technically unfeasible to correct for these effects during personal monitoring. The active flow configuration also allows collection of the sampled particles on a filter for subsequent gravimetric or chemical analysis. The particle mass collected on this back-up filter compared very well with Federal Reference Methods, and can be used as a secondary check on the continuous light-scattering data.

Although the current experiments were conducted on a laboratory bench and sampled outdoor aerosols, there are reasons to believe that the results of this study will also apply to the performance of the pDR when deployed as a wearable monitor. The personal pumps and the pDRs are rugged and not affected by abrupt motion. The use of the $PM_{2.5}$ cyclone in conjunction with active flow prevents larger dust particles, sometimes found indoors and in personal environments, from entering the chamber, settling on the optics, and biasing the results. In conjunction with a wearable humidity monitor and recorder, the active pDRs should prove useful in future exposure assessment and environmental health panel studies.

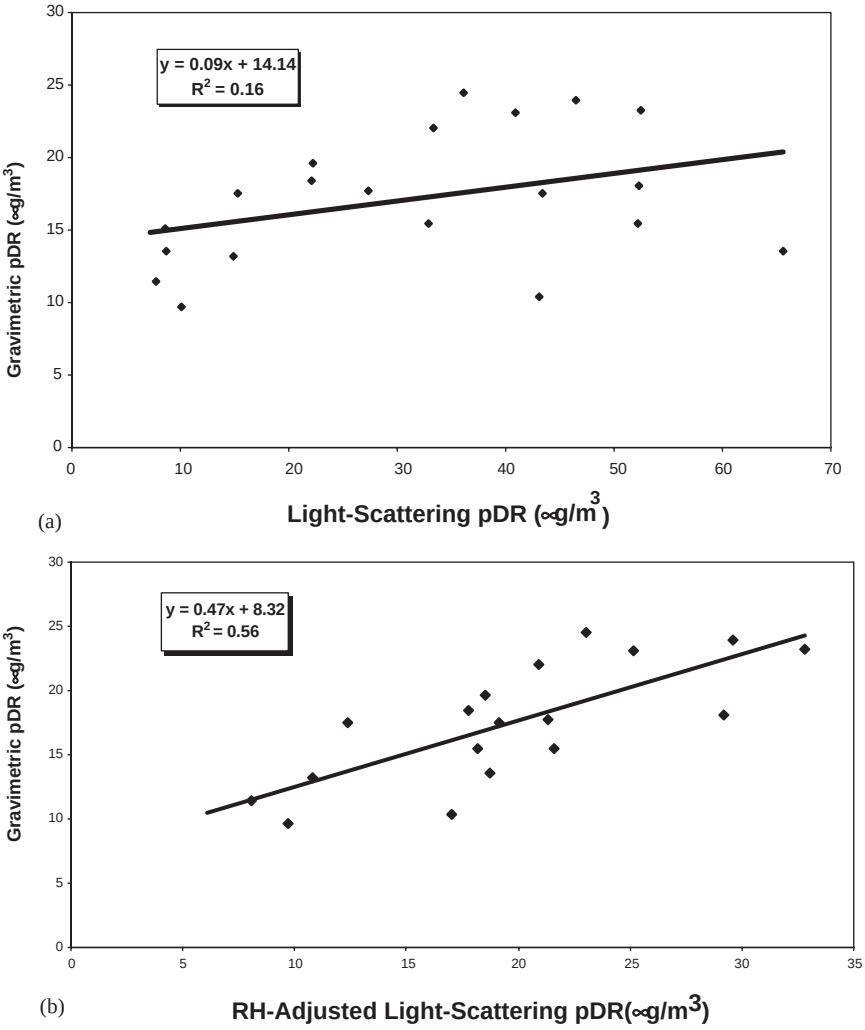



Fig. 9. Correlation between (a) 24-h averaged light-scattering and gravimetric pDR measurements and (b) 24-h averaged RH-adjusted light-scattering and gravimetric pDR measurements.

## Acknowledgements

This work was supported in part by the Southern California Particle Center and Supersite (SCPCS), funded by the US EPA under the STAR program, and the National Institute of Environmental Health Sciences of the National Institute of Health, Grant # ES-11615. Although the research described in this article has been funded in part by the United States Environmental Protection Agency through Grants # 53-4507-0482 and #53-4507-7721 to the University of Southern California, it has not been subjected to the Agency's required peer and policy review and therefore does not necessarily reflect the views of the Agency and no official endorsement should be inferred.

## References

Adams, H.S., Kenny, L.C., Nieuwenhuijsen, M.J., 2001. Design and validation of a high-flow personal sampler for $PM_{2.5}$. Journal of Exposure Analysis and Environmental Epidemiology 11 (1), 5–11.

Buckley, T.J., Waldman, J.M., Freeman, N.G., Lioy, P.J., Marple, V.A., Turner, W.A., 1991. Calibration, intersampler comparison, and field application of a new $PM_{10}$ personal air sampling impactor. Aerosol Science and Technology 14, 380–387.

Chung, A., Chang, D.P.Y., Kleeman, M.J., Perry, K.D., Cahill, T.A., McDougall, E., Stroud, K., 2001. Comparison of real-time instruments used to monitor airborne particulate matter. Journal of the Air and Waste Management Association 51, 109–120.

Clarke, R.W., Coull, B.A., Reinisch, U., Catalano, P., Killingsworth, C.R., Koutrakis, P., Kavouras, I., Lawrence, J., Lovett, E.G., Wolfson, J.M., Verrier, R.L., Godleski, J.J., 2000. Inhaled concentrated ambient particles are associated with hematologic and bronchoalveolar lavage changes in canines. Environmental Health Perspectives 108 (12), 1179–1187.

Day, D.E., Malm, W.C., Kreidenweis, S.M., 2000. Aerosol light scattering measurements as a function of relative humidity. Journal of the Air and Waste Management Association 50, 710–716.

Delfino, R.J., Zeiger, R.S., Seltzer, J.M., Street, D.H., 1998. Symptoms in pediatric asthmatics and air pollution: differences in effects by symptom severity, anti-inflammatory medication use, and particulate averaging time. Environmental Health Perspectives 106, 751–761.

Delfino, R.J., Zeiger, R.S., Seltzer, J.M., Street, D.H., McLaren, C., 2002. Association of asthma symptoms with peak particulate air pollution and effect modification by anti-inflammatory medication use. Environmental Health Perspectives 110, A607–A617.

Dockery, D.W., Pope III, C.A., Xu, X., Spengler, J.D., Ware, J.H., Fay, M.E., Ferris, B.G., Speizer, F.E., 1993. An association between air pollution and mortality in six US cities. New England Journal of Medicine 329, 1753–1759.

Godleski, J.J., Verrier, R.L., Koutrakis, P., Catalano, P., 1999. Mechanisms of morbidity and mortality from exposure to ambient air particles. Health Effects Institute, Investigators Report No. 91, Cambridge, MA.

Gorner, P., Berner, D., Fabries, J.F., 1995. Photometer measurement of polydisperse aerosols. Journal of Aerosol Science 26, 1282–1302.

Katsouyanni, K., Zmirou, D., Spix, C., 1995. Short-term effects of air-pollution on health—a European approach using epidemiological time-series data-the APHEA project- background, objectives, design. European Respiratory Journal 8 (6), 1030–1038.

Kerker, M., 1969. The Scattering of Light. Academic Press, New York.

Koistinen, K.J., Kousa, A., Tenhola, V., Hänninen, O., Oglesby, L., Künzli, N., Georgoulis, L., Jantunen, M.J., 1999. Fine particle ($PM_{2.5}$) measurement methodology, quality assurance procedures and pilot results of the EXPOLIS study. Journal of the Air and Waste Management Association 49, 1212–1220.

Laulainen, N.S., 1993. Summery of conclusions and recommendations from a visibility science workshop; technical basis and issues for a national assessment for visibility impairment, Prepared for US DOE. Pacific Northwest Laboratory PNL-8606.

Liu, L.J.S., Slaughter, J.C., Larson, T.V., 2002. Comparison of light scattering devices and impactors for particulate measurements in indoor, outdoor, and personal environments. Environmental Science and Technology 36 (13), 2977–2986.

Lowenthal, H.D., Rogers, F.C., Saxena, P., Watson, J.G., Chow, J.C., 1995. Sensitivity of estimated light extinction coefficients to model assumptions and measurement errors. Atmospheric Environment 29, 751–766.

Mark, D., Vincent, J.H., Stevens, D.C., 1986. Investigation of the entry characteristics of dust samplers of a type used in the British nuclear industry. Atmospheric Environment 20 (12), 2389–2396.

McMurry, P.H., Zhang, X., Lee, Q.T., 1996. Issues in aerosol measurement for optical assessments. Journal of Geophysical Research 101, 19188–19197.

Michaels, R.A., 1996. Airborne particle excursions contributing to daily average particle levels may be managed via a 1-hr standard with a possible health benefits. Aerosol Science and Technology 25, 436–444.

Michaels, R.A., 1997. Particulate matter policy. Science 278, 1–696.

Michaels, R.A., Kleinman, M.T., 2000. Incidence and apparent health significance of brief airborne particle excursions. Aerosol Science and Technology 32, 93–105.

Misra, C., Singh, M., Shen, S., Sioutas, C., Peter, M.H., 2002. Development and evaluation of a personal cascade impactor sampler (PCIS). Atmospheric Environment 33, 1027–1047.

Ozkaynak, H., Thurston, G.D., 1987. Associations between 1980 US mortality rates and alternative measures of airborne particle concentration. Risk Analysis 7 (4), 449–461.

Patashnick, H., Rupprecht, G., Ambs, J.L., 2001. Development of a reference standard for particulate mater mass in ambient air. Aerosol Science and Technology 34 (1), 42–45.

Peters, A., Frohlich, M., Doring, A., 2001. Particulate air pollution is associated with an acute phase response in men-Results from the MONICA-Augsburg study. European Heart Journal 22 (14), 1198–1204.

Pope III, C.A., Dockery, D.W., Spengler, J.D., Raizenne, M.E., 1991. Respiratory health and $PM_{10}$ pollution. A daily time series analysis. American Review of Respiratory Disease 144(3), 668–674.

Pope, C.A., Thun, M.J., Namboodiri, M.M., Dockery, D.W., Evans, J.S., Speizer, F.E., Heath Jr., C.W., 1995. Particulate air pollution as a predictor of mortality in a prospective study of US adults. American Journal of Respiratory and Critical Care. Medicine 151, 669–674.

Schwartz, J., Dockery, D.W., 1992. Particulate air pollution and daily mortality in Steubenville, Ohio. American Journal of Epidemiology 135 (1), 12–19.

Schwartz, J., Neas, L.M., 2000. Fine particles are more strongly associated than coarse particles with acute respiratory health effects in school children. Epidemiology 11 (1), 6–10.

Schwartz, J., Laden, F., Zanobetti, A., 2002. The concentration–response relation between PM2.5 and daily deaths. Environmental Health Perspectives 110 (10), 1025–1029.

Sioutas, C., Kim, S., Chang, M.C., Terrell, L.L., Gong, H., 2000. Field evaluation of a modified dataram mie scattering monitor for real-time PM2.5 mass concentration measurements. Atmospheric Environment 34, 4829–4838.

Sloane, C.S., 1984. Optical properties of aerosols of mixed composition. Atmospheric Environment 18, 871–878.

Suh, H.H., Koutrakis, P., Spengler, J.D., 1994. Interactions between aerosol acidity and ammonia in indoor environments. Journal of Exposure Analysis and Environmental Epidemiology 4, 1.

Thomas, A., Gebhart, J., 1994. Correlations between gravimetry and light-scattering photometry for atmospheric aerosols. Atmospheric Environment 28 (5), 935–938.

Thomas, K.W., Pellizzari, E.D., Clayton, C.A., Whitaker, D.A., Shores, R.C., Spengler, J., Ozkaynak, H., Frohlich, S.E., Wallace, L.A., 1993. Particle total exposure assessment methodology (PTEAM) 1990 study: method performance for personal, indoor, and outdoor monitoring. Journal of Exposure Analysis and Environmental Epidemiology 3 (2), 203–226.

US EPA, 1996. Air quality criteria for particulate matter, EPA/600/P-95/001aF: US EPA, Cincinnati, OH.